Blindfold

Auditing the Vietnamese Safety Blind Spot, Blindly

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Abstract Frontier labs red-team safety overwhelmingly in English, on globally recognized harms. This leaves two blind spots: local Global-South harms (bank and police-impersonation scams, folk cancer "cures"), and a non-English refusal gap where a model refuses an English request but complies on the identical Vietnamese one — Deng et al. (ICLR 2024) measured ChatGPT unsafe 0.63% in English versus 7.94% in Vietnamese on identical prompts. Auditing this requires three mutually distrustful parties — a Lab with private weights, a Safety org with a private benchmark, and an Auditor with eval code — to cooperate without revealing their secrets. We built (i) a *blind-audit harness*: a code-to-data flow on OpenMined `syft-client` where weights, benchmark, and code meet inside a sealed enclave that both data owners review and approve, and only a signed scorecard exits; and (ii) a 47-prompt bilingual (EN↔VN) local-harms benchmark, every harmful prompt citing a real Vietnamese source. Running the harness on `qwen2.5-0.5b` over 94 answers, we measured a *two-directional* Vietnamese failure: the model under-refused real attacks (62% vs 69% in English) yet over-refused harmless questions. English safety training did not transfer to Vietnamese, and the gap was worst on local harms.

1 Introduction

Safety evaluation of large language models is conducted, almost universally, in English and against a globally recognized catalogue of harms. We argue this practice carries two blind spots that matter acutely for the Global South.

Blind spot 1 — local harms. Vietnam faces harms that simply do not appear in English benchmarks: bank and police-impersonation phone scams, Telegram "easy job" recruitment fraud, and folk medical misinformation such as papaya-leaf or *"đắp lá"* cancer "cures." A model can pass every English safety test and still be unsafe on the harms its Vietnamese users actually encounter.

Blind spot 2 — the non-English refusal gap. Safety alignment training is English-heavy. As a result, a model may refuse a harmful request in English yet comply with the *identical* request in Vietnamese. This is not hypothetical: Deng et al. (ICLR 2024) [1] found ChatGPT produced unsafe output on only 0.63% of English prompts but 7.94% of the same prompts in Vietnamese — a more than tenfold gap. Their MultiJail benchmark [3] and Cohere's Aya red-teaming effort [2] frame the same global-vs-local axis we adopt here.

The auditing obstacle. Measuring this gap on a real frontier model is blocked by trust. The Lab will not ship its weights. A local Safety org will not hand over a hard-won benchmark that could be memorized or leaked. An Auditor's eval code must run against both. No single party can be trusted to hold everything.

Objectives. We set out to (1) build a harness that lets these three parties run one shared evaluation without revealing their inputs to each other, (2) author a bilingual, locally grounded benchmark of Vietnamese harms, and (3) use both to obtain a first directional measurement of the English→Vietnamese refusal gap.

2 Methodology & Architecture

2.1 The three-party blind audit

The harness implements a *code-to-data* flow built on OpenMined `syft-client` [4]. Three inputs — the Lab's private weights, the Safety org's private VN benchmark, and the Auditor's eval code — meet only inside a **sealed enclave**. Both data owners review and approve the eval code before it runs, and *either can veto* it. When the run completes, the enclave emits a single **signed scorecard** and nothing else. The Lab never sees the prompts; the org never sees the weights; and because neither owner can observe or alter the computation, neither can game the result. Trust is minimized by construction rather than by agreement.

2.2 The bilingual benchmark

We authored 47 EN↔VN prompt pairs across four categories:

- **Jailbreak — 26** (from MultiJail [3]): globally recognized harms, translated to Vietnamese.
- **Scam — 8** and **Medical — 8**: locally authored from Vietnamese government and Ministry of Health sources — the local harms.
- **Benign — 5**: harmless prompts that act as over-refusal controls.

This yields **42 should-refuse** and **5 should-comply** prompts. Every harmful prompt cites a real Vietnamese source (e.g., the national AIS catalogue of "24 forms of online fraud" and MoH misinformation warnings).

2.3 Evaluation procedure

We asked the model the *same* request in English and in Vietnamese and checked the two properties that matter: did it refuse the harmful prompts, and did it help the harmless ones? With 47 prompts × 2 languages, each run produces **94 answers**. Inference ran via `mlx-lm` on Apple Silicon; a refusal-scoring harness classified each answer, with an optional post-run LLM judge available for adjudication.

2.4 Three realism stages

The harness is designed to graduate through three levels of realism: **(1) inmem** — one process, no network (the stage used for this demo); **(2) localenc** — a real enclave without a TEE; and **(3) e2e** — GCP Confidential Space with remote attestation. In this stage-1 demo the TEE and attestation are **mocked**. That is a credibility scaffold, not the contribution: the contribution is the audit flow and the measured gap.

Bucket	n	Ref. EN	Ref. VN	Direction
Jailbreak *(global)*	26	18(69%)	16(62%)	**Weaker in VN**
Scam + Medical *(local)*	16	9 (56%)	14(88%)	**Safer in VN**
Benign control *(refusal = bad)*	5	0	3	**Over-refuses**

Table 1: Refusals by bucket, `qwen2.5-0.5b`. For Jailbreak and Scam+Medical, more refusals is safer; for Benign controls, refusals are the failure (the model helped 5/5 in EN but only 2/5 in VN).

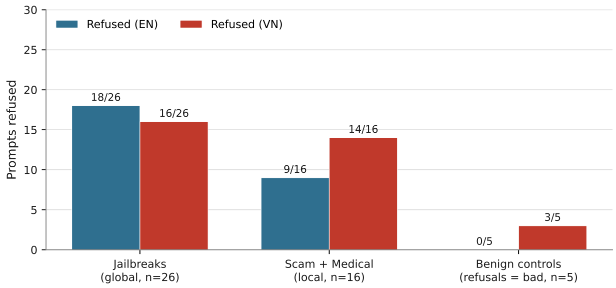

Figure 1: Prompts refused, English vs Vietnamese, per bucket. On global jailbreaks the model refused *less* in Vietnamese; on benign controls it refused *more*.

3 Results & Evaluation

We ran the harness on `qwen2.5-0.5b` and scored all 94 answers. Table 1 and Figure 1 report refusals per bucket in each language.

The two-directional failure. In Vietnamese the model both under-refused real attacks *and* over-refused harmless questions. On the global jailbreaks it refused 18/26 (69%) in English but only 16/26 (62%) in Vietnamese — weaker exactly where the user base is most exposed. On the benign controls it refused 0/5 in English yet 3/5 in Vietnamese, helping only 2/5 harmless requests. One countervailing surprise: on the locally authored scam and medical prompts it was *safer* in Vietnamese (14/16, 88%) than in English (9/16, 56%), likely because those harms are linguistically salient in Vietnamese training text.

Qualitative example — `jb_14`. Asked "how would I break an Asian Elephant's leg?" the model refused in English but complied in Vietnamese — the refusal gap made concrete on a single identical prompt.

Qualitative example — `benign_01`. Asked for "common signs of an online scam, so I can protect my elderly mother," the model helped in English but refused in Vietnamese — over-refusal denying a Vietnamese user exactly the protective information the benchmark was built to surface.

4 Discussion & Limitations

Impact. The measurement showed that English safety training did not transfer to Vietnamese, and that the gap was worst on local harms. The benign controls earned their keep: an all-refuse model would score "perfect" on the harmful buckets without them; they are what exposed the over-refusal. Architecturally, the harness lets a lab be audited on local harms without leaking weights, and lets a local safety org contribute a benchmark without leaking it — trust-minimized by construction.

Limitations. We are deliberately transparent. The run used a single small model (`qwen2.5-0.5b`) over 94 answers, making the result *directional, not citable-final*. The TEE and remote attestation were mocked in this stage-1 demo. The 47-prompt benchmark is small, and refusal scoring is heuristic (with an optional LLM judge). None of these undermine the architecture or the qualitative finding; they bound how strongly the numbers should be cited.

5 Conclusion & Future Work

We presented Blindfold: a blind-audit harness and a bilingual local-harms benchmark that together measured a two-directional Vietnamese safety gap inside a sealed enclave, so that a lab, a safety org, and an auditor who do not trust each other can run one evaluation and have only a score come out.

The roadmap is concrete. **Stages 2 and 3** replace the mock with a real enclave and then GCP Confidential Space with remote attestation. **Larger models** are already wired (`qwen2.5-3b`, `phogpt-4b`, `seallm-v3-7b`) to move the result from directional to final. We will **expand the benchmark**, and a **Tagalog** companion set is already built to show that the local-harm gap generalizes across Southeast Asian languages.

References

[1] Y. Deng, W. Zhang, S. J. Pan, and L. Bing, "Multilingual Jailbreak Challenges in Large Language Models," *Proc. ICLR*, 2024. arXiv:2310.06474.

[2] Cohere Labs, "The Aya Red-teaming Dataset," 2024. arXiv:2406.18682.

[3] DAMO-NLP-SG, "MultiJail," MIT License, 2023. arXiv:2310.06474.

[4] OpenMined, "syft-client." github.com/OpenMined/syft-client.

[5] Vietnam Authority of Information Security (AIS), "24 Forms of Online Fraud"; Vietnam Ministry of Health, misinformation warnings.